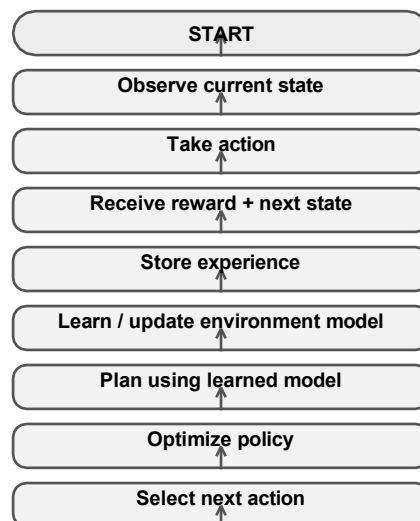


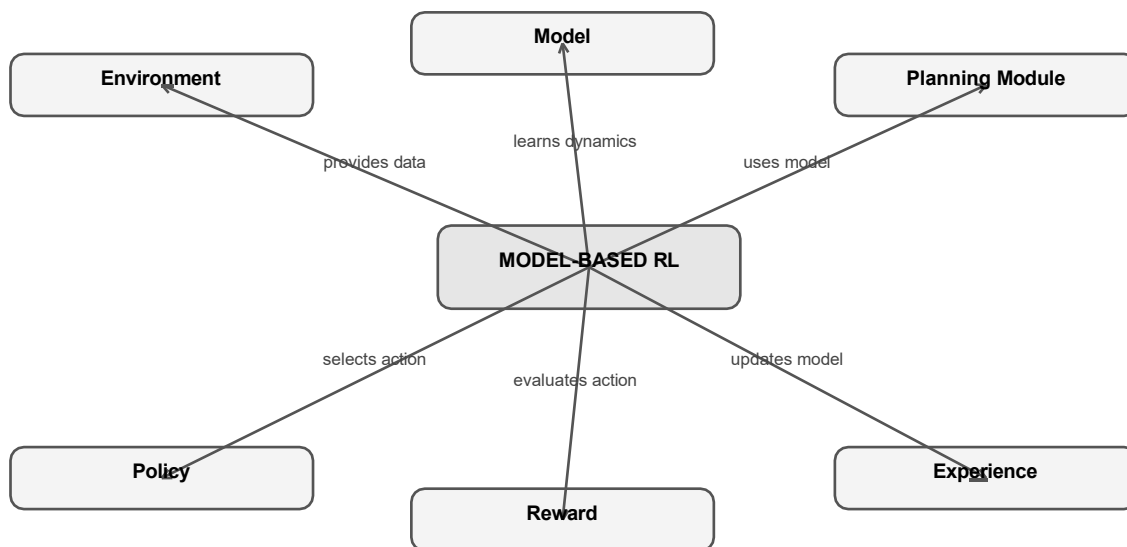
Question 1 — FLOWCHART

Draw a flowchart illustrating the complete workflow of a Model-Based Reinforcement Learning agent from environment interaction to policy optimization.



Question 2 — CONCEPT MAP

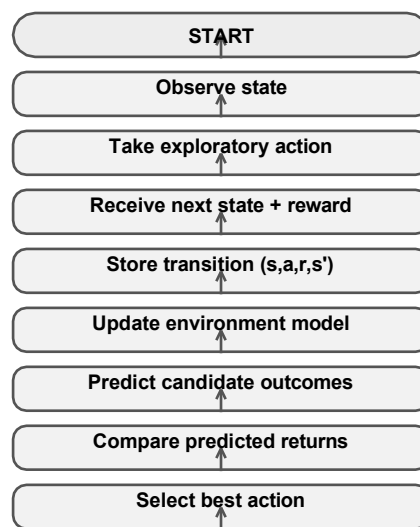
Draw a concept map showing the relationship between the Environment, Model, Planning Module, Policy, and Reward in Model-Based Reinforcement Learning.



Answer: The environment supplies experience and rewards. The model learns environment behavior; planning uses the model; the policy chooses actions; rewards indicate how good the actions are.

Question 3 — FLOWCHART

Design a flowchart explaining how an RL agent learns an environment model before selecting an optimal action.



Question 4 — BLOCK DIAGRAM / ARCHITECTURE

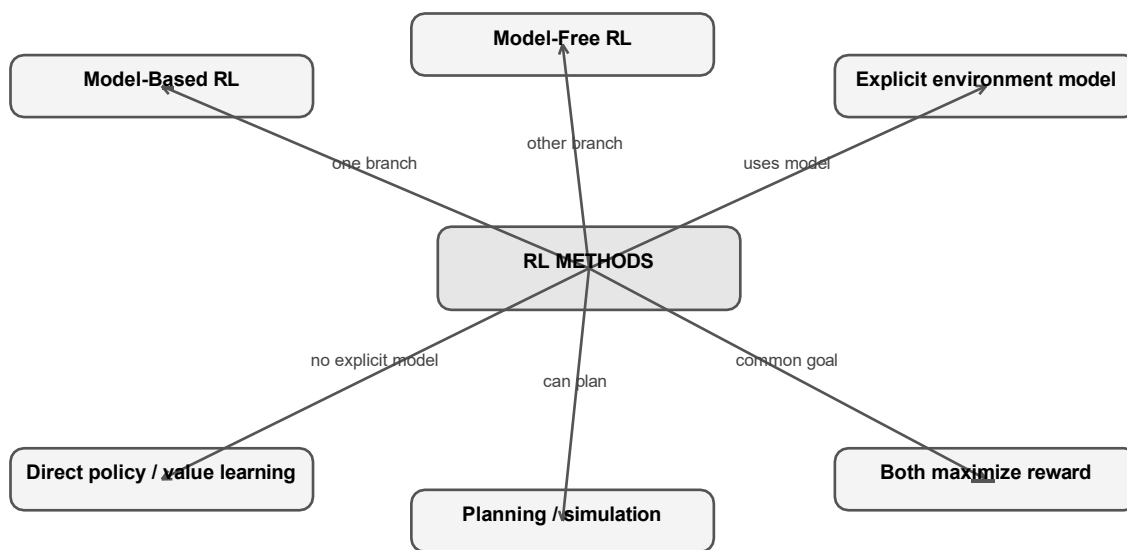
Draw the architecture of a Model-Based RL system showing model learning, planning, and execution components.



Answer: The architecture separates model learning, planning, and action execution. Experience flows from the environment into model learning and then to planning.

Question 5 — CONCEPT MAP

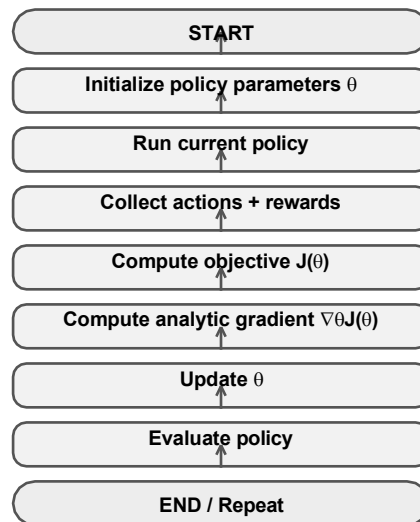
Prepare a concept map comparing Model-Based Reinforcement Learning and Model-Free Reinforcement Learning.



Answer: Model-Based RL uses an environment model and planning. Model-Free RL learns a policy or value function directly from experience. Both aim to maximize reward.

Question 6 — FLOWCHART

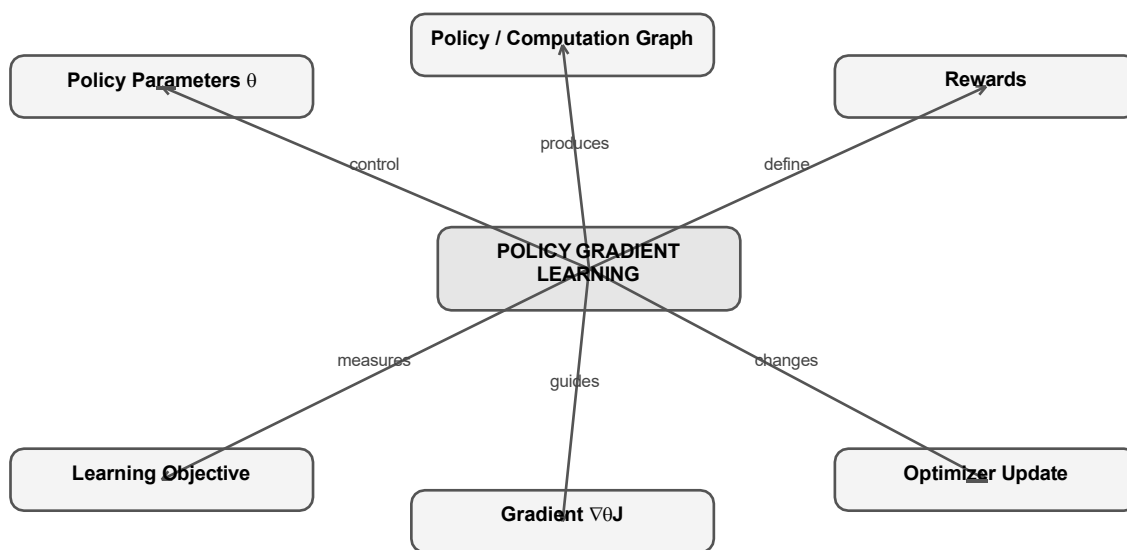
Draw a flowchart explaining the steps involved in analytic gradient computation for policy optimization.



Answer: The policy objective is differentiated analytically to obtain a gradient, and the gradient is used to update policy parameters.

Question 7 — CONCEPT MAP

Design a concept map showing how gradients are computed and propagated during policy learning.



Answer: Policy parameters determine actions. Rewards define the objective, gradients indicate useful parameter changes, and the optimizer applies those changes.

Question 8 — BLOCK DIAGRAM / ARCHITECTURE

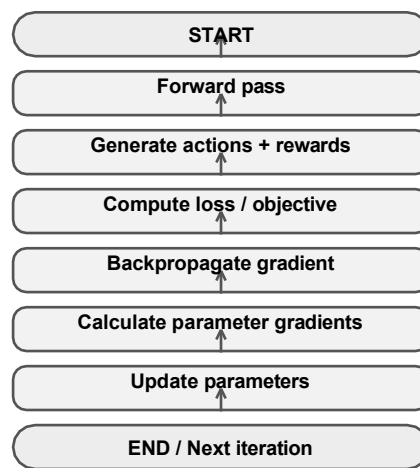
Draw a block diagram illustrating gradient computation, parameter updates, and reward optimization in Reinforcement Learning.



Answer: The reward signal is used to compute a gradient, which updates the policy parameters and improves expected reward.

Question 9 — FLOWCHART

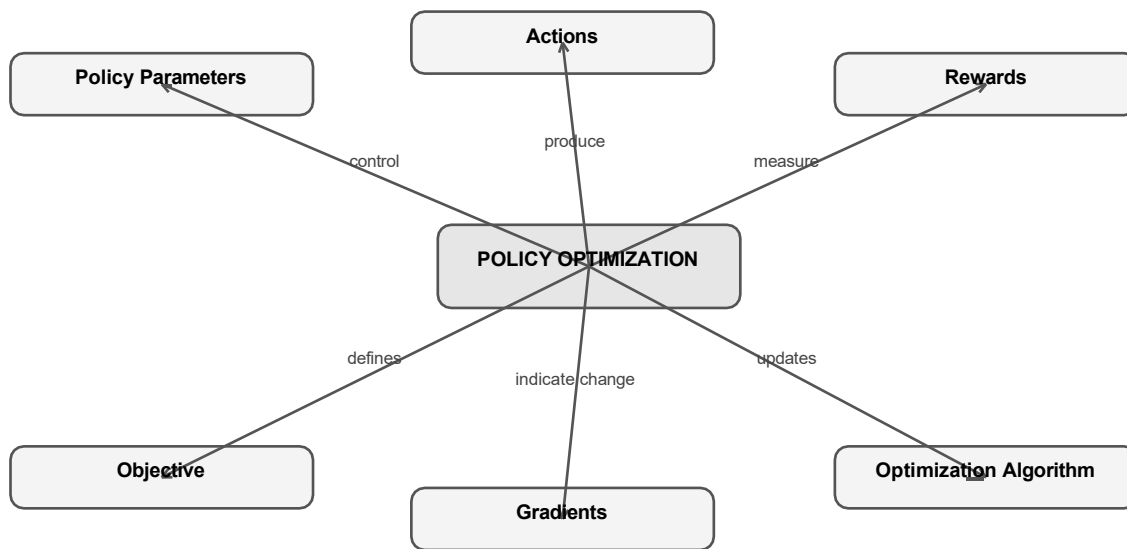
Prepare a flowchart explaining gradient-based optimization using backpropagation in Reinforcement Learning.



Answer: Backpropagation propagates the learning signal backward through a differentiable computation graph to obtain parameter gradients.

Question 10 — CONCEPT MAP

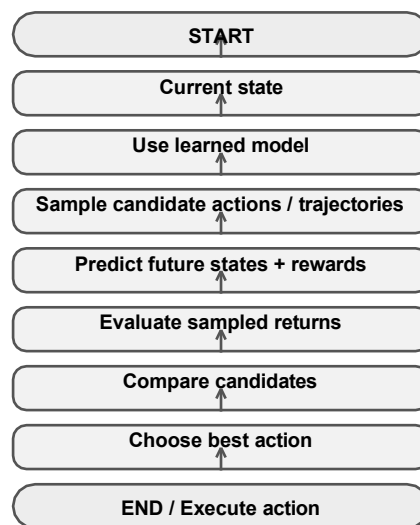
Draw a concept map showing the relationship among policy parameters, gradients, optimization algorithms, and rewards.



Answer: Parameters control actions, actions affect rewards, rewards define the objective, gradients show how parameters should change, and the optimizer performs the update.

Question 11 — FLOWCHART

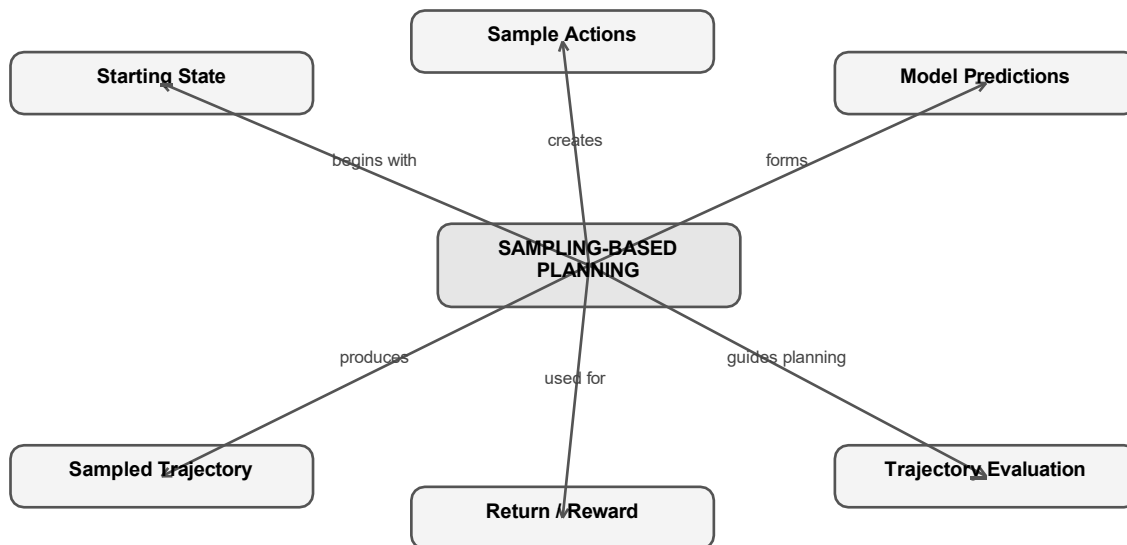
Draw a flowchart illustrating the sampling-based planning process in Model-Based Reinforcement Learning.



Answer: The planner samples possible future trajectories from the learned model, evaluates them, and chooses a good action.

Question 12 — CONCEPT MAP

Prepare a concept map showing how sampled trajectories are generated and evaluated for planning.



Answer: Sampled trajectories are generated from a starting state using sampled actions and model predictions; their returns are evaluated to guide planning.

Question 13 — BLOCK DIAGRAM / ARCHITECTURE

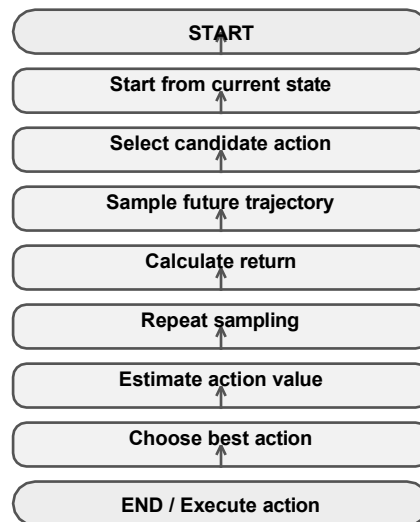
Draw a block diagram showing the interaction between the environment model, planner, sampled actions, and optimal policy.



Answer: The planner uses the environment model to test sampled actions, evaluate predicted outcomes, and select the action that gives the best predicted result.

Question 14 — FLOWCHART

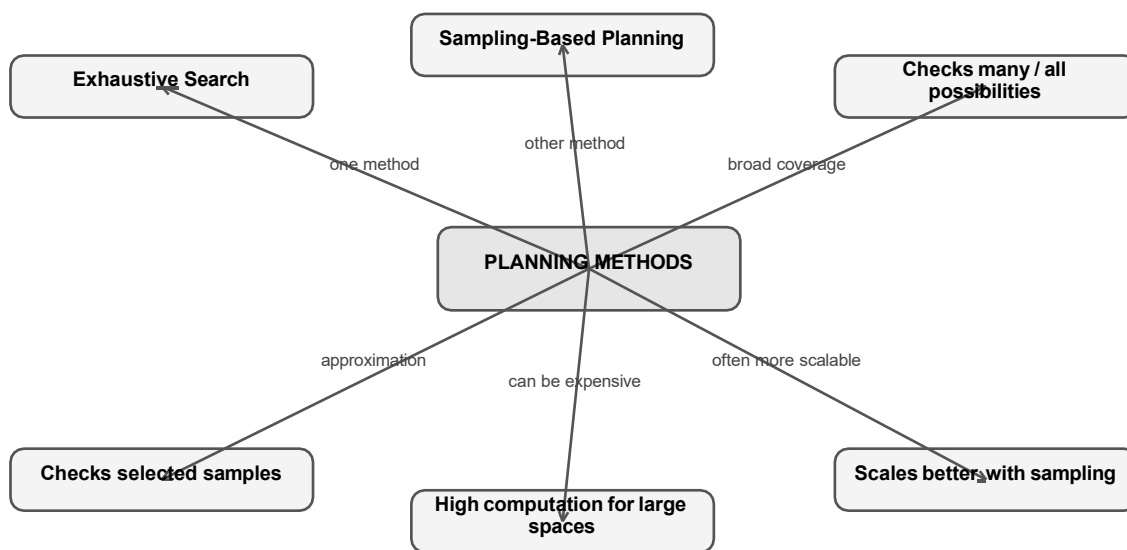
Design a flowchart explaining Monte Carlo sampling for planning in Reinforcement Learning.



Answer: Monte Carlo planning repeatedly samples future trajectories and uses their returns to estimate action quality.

Question 15 — CONCEPT MAP

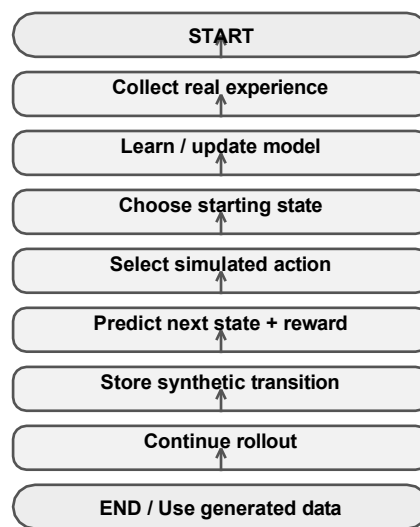
Draw a concept map comparing exhaustive search planning and sampling-based planning techniques.



Answer: Exhaustive search examines a broad set of possibilities but can be expensive. Sampling-based planning checks selected possibilities and is often more scalable.

Question 16 — FLOWCHART

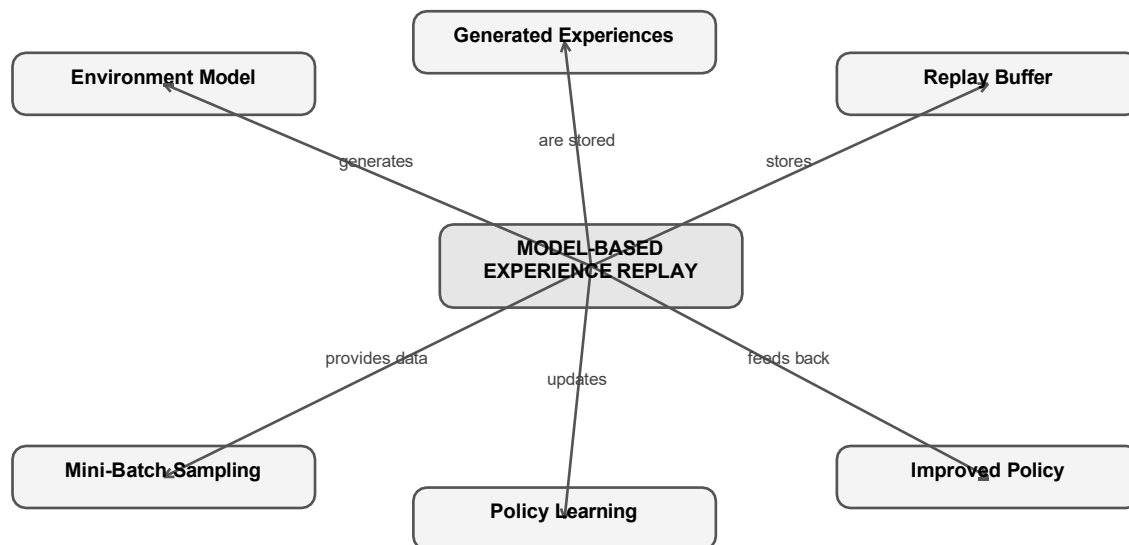
Draw a flowchart illustrating synthetic data generation using a learned environment model.



Answer: A learned model can simulate transitions and generate additional training experiences without requiring every transition from the real environment.

Question 17 — CONCEPT MAP

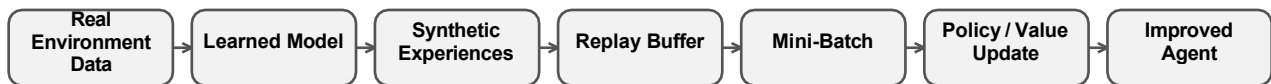
Prepare a concept map showing the relationship among environment model, generated experiences, replay buffer, and policy learning.



Answer: The environment model creates synthetic experiences, the replay buffer stores them, and sampled data is used for policy learning.

Question 18 — BLOCK DIAGRAM / ARCHITECTURE

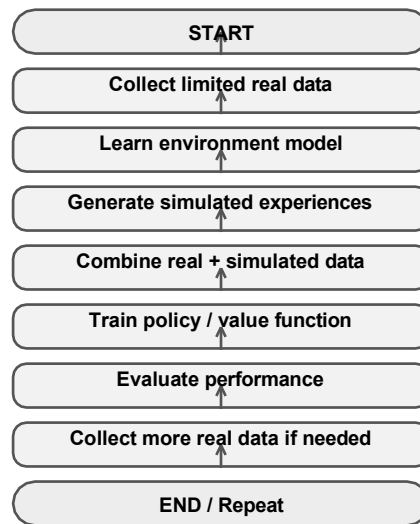
Draw a block diagram explaining model-generated experience replay in Reinforcement Learning.



Answer: Real data is used to learn a model, the model generates synthetic experiences, and replayed mini-batches update the agent.

Question 19 — FLOWCHART

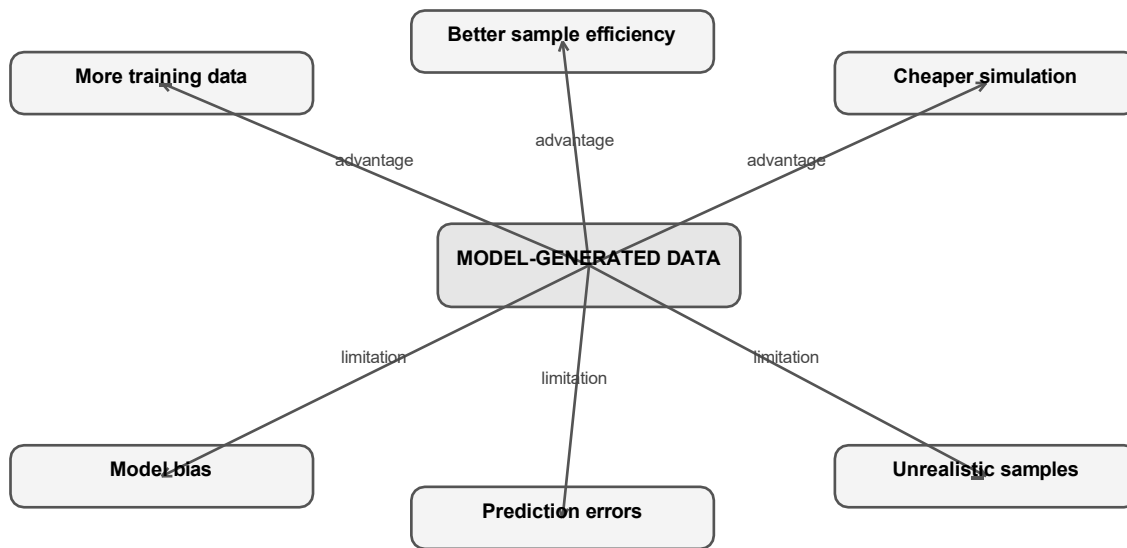
Design a flowchart showing how simulated experiences improve sample efficiency during training.



Answer: Simulated experiences add training samples, allowing learning from more transitions while reducing dependence on real environment interactions.

Question 20 — CONCEPT MAP

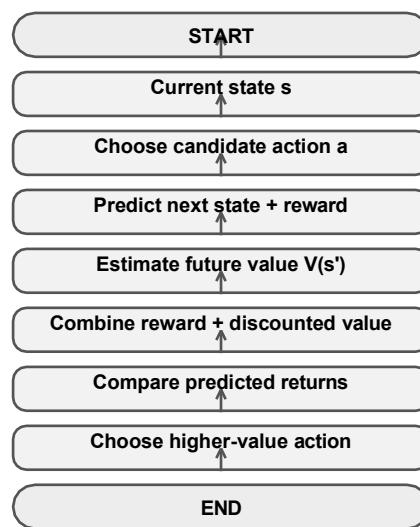
Draw a concept map illustrating the advantages and limitations of model-generated training data.



Answer: Generated data can improve sample efficiency and reduce real interactions, but model bias and prediction errors can make it inaccurate.

Question 21 — FLOWCHART

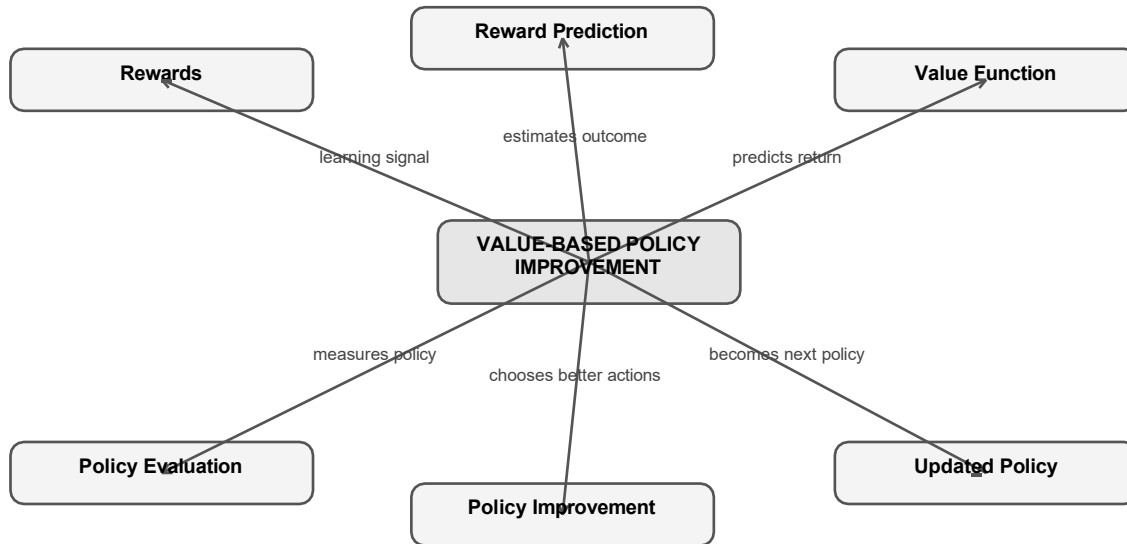
Draw a flowchart explaining value-equivalence prediction for estimating future rewards.



Answer: The model predicts immediate reward and the value function estimates future return, allowing candidate actions to be compared.

Question 22 — CONCEPT MAP

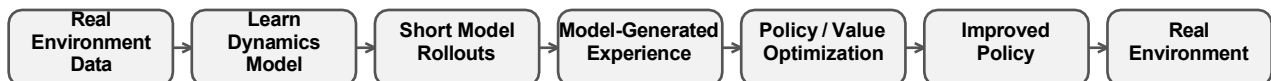
Prepare a concept map showing the relationship among value functions, reward prediction, policy evaluation, and policy improvement.



Answer: Value functions estimate future returns; policy evaluation measures the current policy, and policy improvement uses those estimates to choose better actions.

Question 23 — BLOCK DIAGRAM / ARCHITECTURE

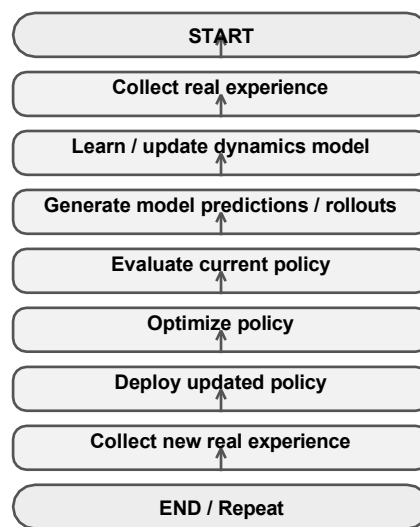
Draw a block diagram illustrating the complete Model-Based Policy Optimization (MBPO) framework.



Answer: MBPO learns dynamics from real data, performs short model rollouts, uses generated experience for optimization, and repeatedly improves the policy.

Question 24 — FLOWCHART

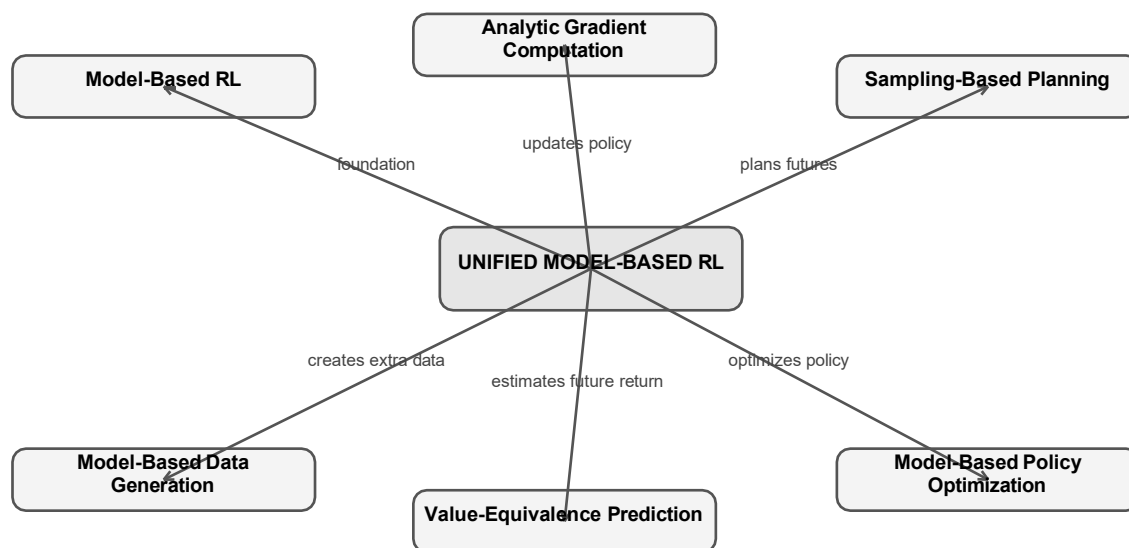
Design a flowchart explaining iterative policy optimization using learned environment dynamics.



Answer: The policy is improved iteratively by learning dynamics, using them for prediction or rollouts, optimizing the policy, and collecting new real data.

Question 25 — CONCEPT MAP

Draw a comprehensive concept map integrating Model-Based RL, Analytic Gradient Computation, Sampling-Based Planning, Model-Based Data Generation, Value-Equivalence Prediction, and Model-Based Policy Optimization into a unified Reinforcement Learning framework.



Answer: The six requested ideas form one loop: learn a model, predict values, plan with samples, generate model data, compute gradients, and optimize the policy.